HNLP Assignment 1: Towards interpretable readmission prediction using DLA on MIMIC-IV dataset

Jasleen Gandhi **Jumanh Atoum** **Zachary Silver**

1 Introduction

Hospital or ICU readmissions, particularly those occurring within 30 days of discharge, remain a persistent challenge in modern healthcare systems. Unplanned readmissions are associated with increased patient morbidity and mortality, substantial financial burden, and fragmented care transitions (Jencks et al., 2009). In the United States, readmission rates are publicly reported and financially penalized under the Hospital Readmissions Reduction Program (HRRP), emphasizing the national priority placed on reducing avoidable hospital or ICU returns (Centers for Medicare & Medicaid Services, 2023). Despite ongoing quality improvement initiatives, accurately identifying patients at high risk for readmission before discharge remains an unresolved problem.

Natural language processing (NLP) approaches have demonstrated improved performance in predicting hospital or ICU readmission compared to models that rely solely on traditional structured variables (Wang et al., 2018; Rajkomar et al., 2018). Nevertheless, many existing prediction models continue to depend heavily on structured electronic health record (EHR) data such as demographics, laboratory values, and comorbidity indices, which, while informative, fail to capture the contextual and semantic nuances embedded in clinical narratives.

To facilitate research in this domain while preserving patient privacy, large-scale deidentified datasets such as MIMIC-IV have been developed (Johnson et al., 2023). This publicly available in-hospital dataset contains detailed clinical information from patients admitted to intensive care units and emergency departments and was curated to balance data accessibility with patient confidentiality. The availability of such datasets is essential for advancing clinical predictive systems and improving patient care, as hospital or ICU readmission prediction remains a persistent challenge despite advances in modeling techniques. Importantly, the richness of data within MIMIC-IV, including both structured and unstructured documentation, enables investigation of diverse outcomes such as hospital or ICU readmission, or absence of return to care.

In this work, we use MIMIC-IV to predict 30-day post-discharge outcomes by integrating clinical notes with structured EHR data. We model whether patients experience no return or hospital or ICU readmission, and apply Differential Language Analysis to identify linguistic patterns associated with each outcome. This approach combines multimodal prediction with interpretable language analysis to better understand how clinical documentation reflects readmission risk and patient acuity.

2 Background

Accurate prediction of hospital or ICU readmission is essential to understand the mechanisms underlying clinical deterioration and failed care transitions. Although many models rely primarily on structured EHR data such as demographics, laboratory values, diagnosis codes, and prior utilization, these variables often lack the contextual information necessary to explain why a patient returns. In contrast, discharge summaries provide a clinically rich synthesis of hospitalization, incorporating provider reasoning, complications, treatment response, residual instability at discharge, follow-up planning, and social and behavioral determinants of health (Meystre et al., 2008). As a semantically dense representation of the patient's transition from acute illness to recovery, discharge notes encode both objective clinical status and subjective assessments of readiness for discharge.

Advances in NLP and deep learning have shown that incorporating discharge narratives significantly improves prediction of 30-day hospital or ICU readmission. Studies using MIMIC-III and MIMIC-IV demonstrate that models leveraging clinical text outperform those trained solely on structured variables, with recurrent, convolutional, and transformer-based architectures capturing meaningful semantic risk patterns (Huang et al., 2019; Liu et al., 2021). However, most existing approaches treat readmission as a single outcome without distinguishing between hospital readmission after discharge to the community and ICU readmission following ICU transfer. This distinction is mechanistically important. Thirty-day hospital readmission is often driven by care transitions, outpatient access, and social complexity, whereas ICU readmission, frequently occurring within 48–72 hours, reflects unresolved physiological instability and is strongly associated with higher in-hospital mortality and prolonged length of stay (Brown et al., 2012). Conflating these outcomes may obscure distinct risk pathways and dilute clinically relevant semantic signals embedded in discharge documentation, limiting both interpretability and clinical utility.

3 Dataset

We use the Medical Information Mart for Intensive Care (MIMIC-IV) database, a large de-identified electronic health record EHR dataset containing detailed emergency department visits and hospital admissions from Beth Israel Deaconess Medical Center between 2008 - 2022. Each entry in the dataset represents one ED or hospital admission. There are 425,087 ED visits and 546,028 hospital visits, for a total of 971,141 visit data points.

Each patient has a patient ID number, so for a given visit, we can check whether the same patient was re-admitted within 30 days. 56% of users were readmitted within 30 days, and 44% were not.

Each visit also contains free-form clinical notes taken by doctors and nurses. These notes are fairly long, containing an average of 2731 tokens, so automated analysis of these notes may prove useful. Our goal is to use the clinical notes to predict whether the patient will be readmitted within 30 days. One difficulty provided by this dataset is that the notes contain a variety of medical abbreviations, numerical data, and misspellings. For example, a portion of one admission's note reads:

```
ADMISSION LABS: ___ 06:39AM BLOOD
WBC-6.9 RBC-3.98* Hgb-14.1 Hct-41.1
MCV-103* MCH-35.4* MCHC-34.3 RDW-15.8*
Plt ___
```

Such sequences interleave medical concepts with numerical values and formatting markers, making token boundaries ambiguous. Standard NLP pipelines often treat tokens as atomic units, remaining largely agnostic to their internal structure (Manning and Schütze, 1999). This limitation is particularly pronounced in clinical text, where meaningful information may be embedded within composite tokens (e.g., lab-value pairs or abbreviated measurements).

These challenges motivate the use of carefully designed preprocessing and representation strategies that balance noise reduction with preservation of clinically meaningful semantics.

4 Data Pre-processing

4.1 Cohort Construction and Unit of Analysis

We defined the unit of analysis as the index hospital admission (`hadm_id`) in the dataset (Johnson, 2023). Admissions were chronologically ordered per patient (`subject_id`), and for each index discharge, we determined whether a subsequent return to acute care occurred within a 30-day window.

Returns were initially categorized as: (1) no return, (2) hospital readmission without ICU stay, or (3) readmission involving an ICU stay. For the purposes of this study, these categories were collapsed into a binary outcome indicating whether any readmission occurred within 30 days.

To prevent information leakage, only encounters occurring after the index discharge were considered when constructing outcome labels.

4.2 Text Construction and Temporal Alignment

For each index admission, we included all clinical notes with timestamps less than or equal to the discharge time. Notes were concatenated at the admission level to form a single document per admission, consistent with prior work treating discharge summaries as compact representations of the hospitalization trajectory.

Structured covariates were aggregated at the admission level, and all time-varying features were restricted to measurements recorded between admission and discharge to ensure temporal validity.

4.3 Clinical Text Preprocessing

Clinical notes in MIMIC-IV are semi-structured and contain both meaningful narrative content and noise such as timestamps, units, and formatting artifacts. To preserve clinically relevant semantics while reducing noise, we applied a minimal preprocessing pipeline:

- Lowercasing all text
- Removing de-identification placeholders (e.g., ___)
- Removing numeric-heavy tokens (e.g., `03am`, `1050ml`, `10mg`)
- Tokenization using a regex-based tokenizer restricted to alphabetic tokens

We intentionally avoided aggressive normalization techniques such as stemming or lemmatization, as these may obscure clinically meaningful distinctions (e.g., *ascites*, *cirrhosis*, *lactulose*). Preserving domain-specific terminology is critical in clinical NLP, where subtle lexical differences may correspond to distinct physiological conditions.

4.4 Label Construction

The classification target was 30-day readmission, defined as a binary variable:

- 1: patient readmitted within 30 days of discharge
- 0: no readmission within 30 days

Labels were constructed at the admission level using `hadm_id`, ensuring a one-to-one correspondence between each admission, its associated discharge documentation, and the outcome. This avoids ambiguity arising from multiple admissions per patient.

5 Method

This work integrates clinical natural language processing with vector semantic representations to model patient readmission risk from unstructured text presented in Table 1. Our approach is grounded in the distributional hypothesis, which posits that linguistic meaning can be inferred from patterns of word usage in context (Harris, 1954; Firth, 1957). In the clinical domain, discharge summaries serve as compressed, expert-curated representations of a patient's hospital trajectory, containing salient diagnostic, procedural, and treatment information relevant to downstream outcomes.

5.1 Word Modeling

We investigate two complementary vector space models that operationalize semantic meaning through co-occurrence statistics and contextual embeddings. Both approaches map documents into a continuous vector space $\mathbf{d}_i \in \mathbb{R}^k$, enabling linear classification.

Word2Vec. We train a neural embedding model using the skip-gram architecture (Mikolov et al., 2013), which learns word representations by predicting surrounding context words. Formally, the model maximizes:

$$\sum_{w \in D} \sum_{c \in \text{context}(w)} \log P(c \mid w)$$

where $P(c \mid w)$ is parameterized via a softmax over word vectors. We use a vector size of 200, window size of 5, and a minimum frequency threshold of 2. Each document is represented by averaging its constituent word embeddings:

$$\mathbf{d}_i = \frac{1}{|W_i|} \sum_{w \in W_i} \mathbf{v}_w$$

This approach captures local contextual semantics and smooths lexical variability, which is particularly important in clinical narratives with heterogeneous phrasing.

Latent Semantic Analysis (LSA). We construct a term-document matrix using TF-IDF weighting and apply truncated Singular Value Decomposition (SVD) to obtain a low-dimensional representation (Deerwester et al., 1990). Specifically, we compute:

$$\mathbf{X} \approx \mathbf{U}_k \mathbf{\Sigma}_k \mathbf{V}_k^\top$$

where $\mathbf{X}$ is the TF-IDF matrix and $k = 200$ is the number of latent components. We use a maximum of 10,000 features, remove both standard English and domain-specific stopwords, and filter terms using `min_df` $= 5$ and `max_df` $= 0.9$. LSA captures global co-occurrence structure and latent clinical topics, providing a complementary representation to Word2Vec.

5.2 Classification Model

We train a logistic regression classifier on top of each representation. Given a document vector $\mathbf{d}_i$, the model estimates:

$$P(y_i = 1 \mid \mathbf{d}_i) = \sigma(\mathbf{w}^\top \mathbf{d}_i + b)$$

where $\sigma(\cdot)$ is the sigmoid function. Logistic regression is chosen for its interpretability and its ability to evaluate the quality of learned representations without introducing additional model complexity.

Method	Representation	Classifier
Word2Vec	Averaged embeddings	Log. Reg.
LSA	SVD topic vectors	Log. Reg.

Table 1: Comparison of representation methods

Model	Accuracy	Precision	Recall	F1
Word2Vec	0.685	0.675	**0.854**	**0.754**
LSA	**0.694**	**0.723**	0.744	0.734

Table 2: Test performance comparison

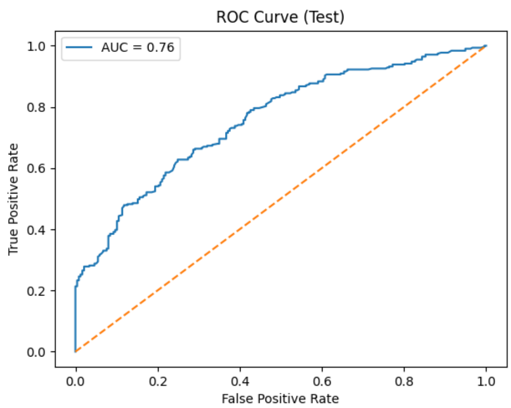

Figure 1: ROC curve for Word2Vec model (AUC = 0.76)

6 Results

6.1 Overall Performance

We evaluate both Word2Vec and LSA representations using logistic regression on a held-out test set. Table 2 summarizes the performance metrics.

LSA achieves slightly higher overall classification performance, with a test AUC of 0.79 compared to 0.76 for Word2Vec. However, Word2Vec demonstrates higher recall (0.85 vs. 0.74), indicating a stronger ability to identify readmitted patients. This reflects a trade-off between sensitivity and precision, where Word2Vec favors detecting positive cases at the expense of increased false positives.

6.2 ROC Analysis

The ROC curves for both models are shown in Figure 2 and 1. Both models significantly outperform random chance, indicating that clinical text contains meaningful predictive signal for readmission.

LSA exhibits a consistently higher true positive rate across most thresholds, suggesting that global co-occurrence patterns captured by latent semantic structure provide a more robust ranking of patient risk.

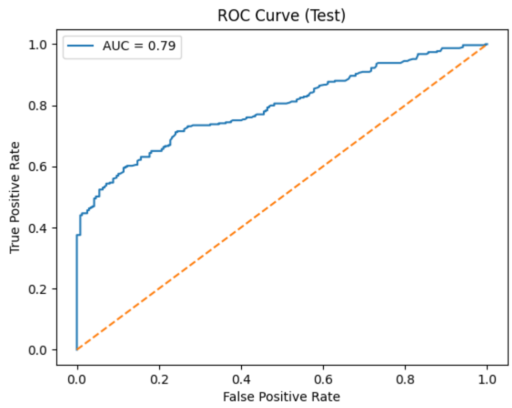

Figure 2: ROC curve for LSA model (AUC = 0.79)

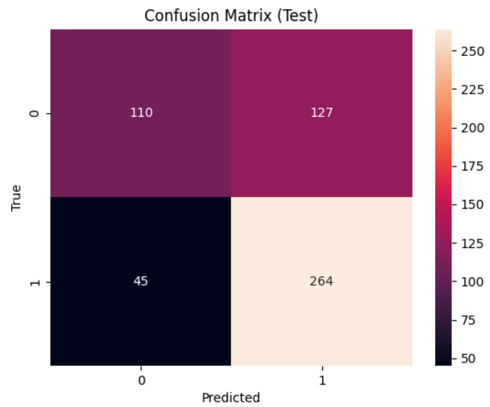

Figure 3: Confusion matrix for Word2Vec

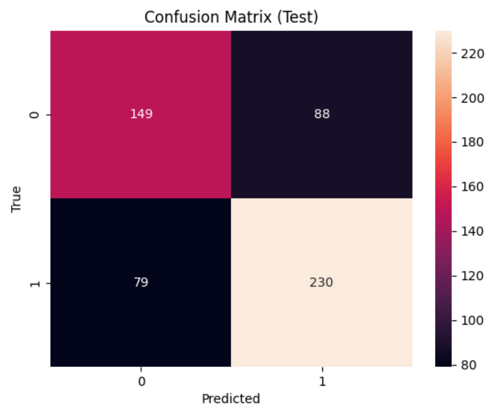

Figure 4: Confusion matrix for LSA

6.3 Error Analysis via Confusion Matrices

Confusion matrices for both models are shown in Figures 3 and 4.

Word2Vec produces more true positives (230) but also more false positives (88), reflecting its higher recall. In contrast, LSA reduces false positives (45) but at the cost of missing more readmission cases. This suggests that Word2Vec captures broader contextual similarity, while LSA provides more discriminative boundaries.

6.4 Semantic Representation Analysis

To better understand the learned representations, we visualize selected word embeddings using PCA in Figure 5.

The embedding space reveals clinically meaningful clustering. For example, medications such as *furosemide*, *lisinopril*, and *atorvastatin* cluster together, reflecting shared therapeutic contexts. Similarly, transplant-related terms (*tacrolimus*, *cyclosporine*) form a distinct group, indicating that the model captures domain-specific semantic relationships. These patterns support the ability of Word2Vec to encode fine-grained contextual similarity in clinical language.

6.5 Lexical Signal and Clinical Relevance

Figure 6 shows the distribution of clinically relevant terms across readmission classes.

Words associated with severe or chronic conditions (e.g., *transplant*, *kidney*, *gout*) appear more frequently in readmitted patients, suggesting that the model leverages clinically meaningful risk indicators. Conversely, routine medications and less acute conditions are more evenly distributed or associated with non-readmission.

This trend is further supported by LSA feature weights, where high-risk terms such as *transplant*, *kidney*, and *cholangitis* contribute positively to readmission prediction, while terms such as *lisinopril*, *atorvastatin*, and *thyroid* are associated with lower risk.

6.6 Interpretation

Overall, the results highlight a key distinction between the two representation approaches. LSA captures global co-occurrence structure, leading to more stable classification performance, while Word2Vec captures local contextual semantics, resulting in higher sensitivity to clinically relevant signals. This trade-off suggests that different vector semantic representations encode complementary aspects of clinical language, which may be beneficial for downstream predictive modeling.

7 Planned Methods

7.1 Approach Overview

We propose a multimodal framework for predicting 30-day hospital or ICU readmission using both unstructured clinical text and structured electronic health record (EHR) data. The core hypothesis is that discharge documentation encodes latent clinical signals related to patient acuity, care transitions, and unresolved physiological instability, which are not fully captured by structured variables alone.

Our approach combines vector semantic representations with interpretable statistical modeling. Specifically, we compare two text representation paradigms grounded in distributional semantics (Harris, 1954): (1) local contextual embeddings using Word2Vec, and (2) global co-occurrence representations using Latent Semantic Analysis (LSA). These representations are used as inputs to a logistic regression classifier, allowing us to evaluate how different semantic abstractions influence predictive performance.

In parallel, we incorporate Differential Language Analysis (DLA) to identify statistically significant linguistic features associated with readmission outcomes. This provides an interpretable layer that complements predictive modeling by highlighting clinically meaningful terms and phrases that differentiate patient trajectories.

7.2 Modeling Framework and Design Choices

The modeling pipeline consists of three main stages: (1) preprocessing and tokenization of clinical notes, (2) transformation into vector semantic representations, and (3) classification using logistic regression. Word2Vec is implemented using the skip-gram architecture with a vector dimension of 200, window size of 5, and minimum token frequency of 2, enabling the model to capture local contextual relationships between clinical terms. LSA is constructed using TF-IDF features (maximum 10,000 terms, `min_df` $= 5$, `max_df` $= 0.9$) followed by truncated singular value decomposition with 200 latent components.

Logistic regression is selected as the classifier due to its interpretability and its ability to isolate the contribution of representation quality without introducing additional model complexity. Hyperparameters of interest include regularization strength, feature dimensionality, and vocabulary thresholds, all of which influence the trade-off between model expressiveness and generalization.

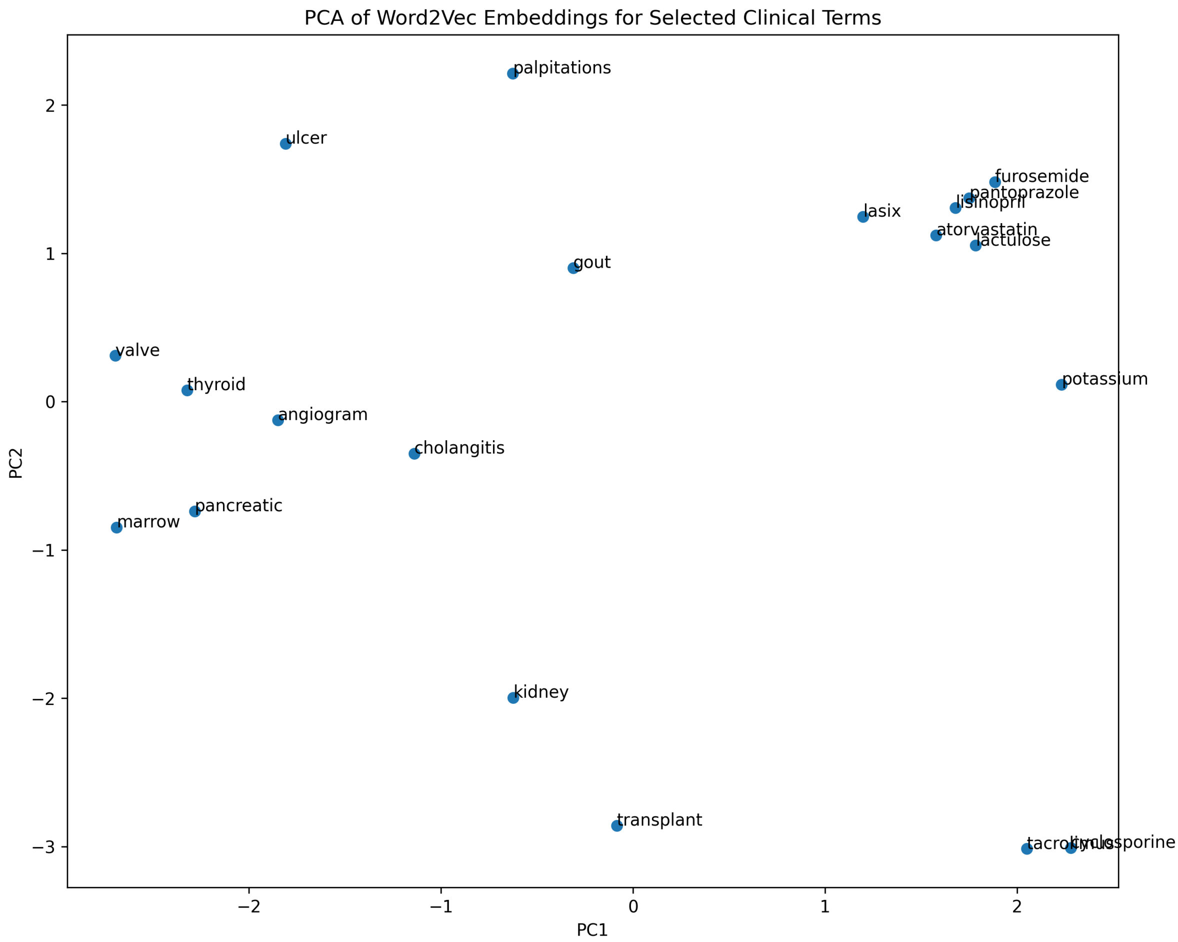

Figure 5: PCA projection of Word2Vec embeddings for selected clinical terms

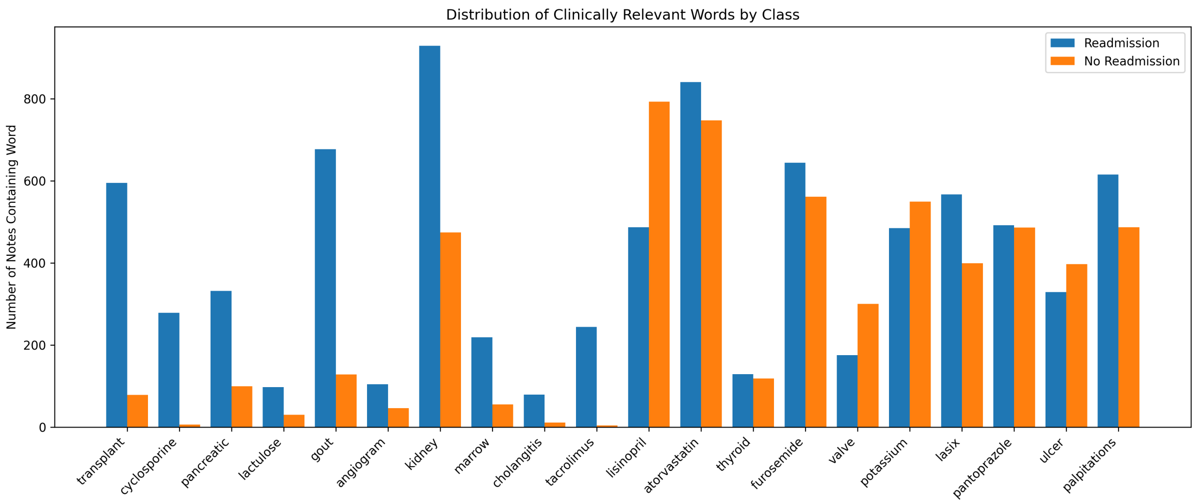

Figure 6: Frequency of selected clinical terms by class

This framework is particularly interesting because it enables a direct comparison between two fundamentally different representations of clinical language: one based on local predictive context (Word2Vec) and one based on global statistical structure (LSA). This allows us to investigate how different semantic assumptions affect both predictive accuracy and interpretability.

Pipeline Overview:
Clinical Notes + Structured EHR → Preprocessing → (Word2Vec / LSA) → Logistic Regression → Prediction
↓
Differential Language Analysis (Interpretability)

Figure 7: Proposed multimodal prediction and interpretation pipeline

Mock ROC Curves:
Word2Vec vs LSA showing separation from random baseline and relative performance differences.

Figure 8: Mockup of ROC curves comparing representation methods

7.3 Interpretability via Differential Language Analysis

To complement predictive modeling, we apply Differential Language Analysis to quantify associations between linguistic features and readmission outcomes. Using regression-based effect sizes and statistical significance testing, we identify terms that are overrepresented in patients who experience hospital or ICU readmission versus those who do not.

This analysis provides insight into the clinical semantics captured by the model, enabling interpretation beyond aggregate performance metrics. For example, we expect terms related to chronic disease burden (e.g., *transplant*, *kidney*) or treatment complexity (e.g., immunosuppressants) to be associated with higher readmission risk, while routine medication terms may correlate with lower risk.

7.4 Proposed Pipeline

8 Result Mockup

8.1 Performance Comparison

We will evaluate model performance using accuracy, precision, recall, F1-score, and AUC. Results will be presented in a comparative table and ROC curves to illustrate the trade-offs between representation methods.

This figure will demonstrate whether the proposed approach effectively captures predictive signal from clinical text. A higher AUC for one representation would indicate superior ranking ability

Mock Feature Importance Plot:
Top positive and negative words associated with readmission risk.

Figure 9: Mockup of clinically relevant terms associated with readmission

in distinguishing patients at risk of hospital or ICU readmission.

8.2 Interpretability and Clinical Insights

To highlight interpretability, we will present key features identified through Differential Language Analysis and compare their distribution across outcome classes.

This figure will provide insight into which clinical concepts drive model predictions. For example, we expect terms related to severe comorbidities, transplant status, and treatment complexity to be associated with higher readmission risk, while routine medication and stable conditions may be associated with lower risk.

8.3 Semantic Representation Analysis

We will also visualize embedding spaces (e.g., PCA projections) to demonstrate how clinical concepts cluster in the learned representation space.

These visualizations will highlight whether semantically related clinical terms group together, providing qualitative evidence that the model captures meaningful structure in clinical language. Together, these figures will form a coherent narrative: (1) the model works, (2) it captures clinically meaningful signal, and (3) it provides interpretable insights into readmission risk.

Model Card: 30-Day Hospital or ICU Readmission Prediction

Model Summary: This model predicts 30-day hospital or ICU readmission using clinical notes and structured EHR data from MIMIC-IV.

Data and Population: The model is trained on the :contentReference[oaicite:0]index=0 dataset, which contains deidentified clinical records from patients admitted to intensive care units and emergency departments at Beth Israel Deaconess Medical Center. The population reflects a tertiary-care academic medical center and may not generalize to other settings.

Input Features: Unstructured clinical notes (discharge summaries and prior notes) and structured EHR variables aggregated at the admission level.

Output: Binary prediction indicating whether a patient will experience hospital or ICU readmission within 30 days.

Modeling Approach: Text is represented using Word2Vec or LSA embeddings and classified using logistic regression. Differential Language Analysis is used to identify interpretable linguistic features associated with readmission risk.

Assumptions:

- Clinical documentation reflects patient acuity and care transitions
- Pre-discharge data is sufficient for prediction
- Readmission labels accurately reflect outcomes

Limitations:

- Single-center dataset limits generalizability
- Documentation bias may influence results
- Temporal dynamics are not explicitly modeled

Intended Use: For research and interpretability of clinical language patterns, not for direct clinical decision-making.

Table 3: Model card for the proposed readmission prediction framework

Statistic	Token Count
Count	331,793
Mean	3,075.51
Std	1,293.24
Min	103
25%	2,158
50% (Median)	2,902
75%	3,774
Max	17,254

Table 4: Summary Statistics for Token Counts

References

Steven E Brown, Sarah J Ratcliffe, Jeremy M Kahn, and Scott D Halpern. 2012. Icu readmissions: epidemiology and outcomes. *Critical Care Medicine*.

Centers for Medicare & Medicaid Services. 2023. Hospital readmissions reduction program (hrrp).

Scott Deerwester, Susan Dumais, George Furnas, Thomas Landauer, and Richard Harshman. 1990. Indexing by latent semantic analysis. *JASIS*.

J.R. Firth. 1957. *A synopsis of linguistic theory 1930–1955*.

Zellig S. Harris. 1954. Distributional structure. *Word*.

Kexin Huang, Jaan Altosaar, and Rajesh Ranganath. 2019. Clinicalbert: Modeling clinical notes and predicting hospital readmission. *arXiv preprint*.

Rank	ICD Code	ICD Ver.	Patient Count	Percent (%)	Disease Description
1	4019	9	102,368	1.61	Unspecified essential hypertension
2	E785	10	84,570	1.33	Hyperlipidemia, unspecified
3	I10	10	83,775	1.32	Essential (primary) hypertension
4	2724	9	67,293	1.06	Other and unspecified hyperlipidemia
5	Z87891	10	62,806	0.99	Personal history of nicotine dependence
6	K219	10	56,157	0.88	Gastro-esophageal reflux disease without esophagitis
7	53081	9	48,628	0.76	Esophageal reflux
8	25000	9	43,077	0.68	Diabetes mellitus without mention of complication, type II or unspecified type, not stated as uncontrolled
9	F329	10	41,876	0.66	Major depressive disorder, single episode, unspecified
10	41401	9	38,957	0.61	Coronary atherosclerosis of native coronary artery

Table 5: Top 10 ICD codes in the MIMIC-IV dataset that were used for [insert task]

Stephen F Jencks, Mark V Williams, and Eric A Coleman. 2009. Rehospitalizations among patients in the medicare fee-for-service program. *New England Journal of Medicine*.

Alistair Johnson, Tom Pollard, Lu Shen, et al. 2023. Mimic-iv, a freely accessible electronic health record dataset. *Scientific Data*.

Alistair et al. Johnson. 2023. Mimic-iv. *Scientific Data*.

Fei Liu et al. 2021. Multimodal ehr-based prediction of patient outcomes. *Journal of Biomedical Informatics*.

Christopher D. Manning and Hinrich Schütze. 1999. *Foundations of Statistical Natural Language Processing*. MIT Press.

Stephane Meystre, Guergana Savova, Karin Kipper-Schuler, and John Hurdle. 2008. Extracting information from textual documents in the electronic health record: a review. *Yearbook of Medical Informatics*.

Tomas Mikolov, Kai Chen, Greg Corrado, and Jeffrey Dean. 2013. Efficient estimation of word representations in vector space. In *ICLR*.

Alvin Rajkomar, Eyal Oren, Kai Chen, et al. 2018. Scalable and accurate deep learning with electronic health records. *npj Digital Medicine*.

Yanshan Wang, Liwei Wang, Majid Rastegar-Mojarad, et al. 2018. Clinical information extraction applications: A literature review. *Journal of Biomedical Informatics*.